Quiz.

1.

(1) $\dfrac{1}{1+4x}$

$= \dfrac{1}{1-(-4x)}$

$= \displaystyle\sum_{n=0}^{\infty} (-4x)^n$

① $x = -\frac{1}{4}$ ② $x = \frac{1}{4}$

$\displaystyle\sum_{n=0}^{\infty} 1 = \infty$ (발산) $\displaystyle\sum_{n=0}^{\infty} (-1)^n =$ 진동

$\therefore R = \frac{1}{4}$, $-\frac{1}{4} < x < \frac{1}{4}$ //

(2)

$\left(\dfrac{1}{1+4x}\right)' = \dfrac{-4}{(1+4x)^2}$

$\therefore \dfrac{x}{(1+4x)^2} = \dfrac{-4}{(1+4x)^2} \cdot \left(-\dfrac{x}{4}\right)$

$\therefore \dfrac{x}{(1+4x)^2} = \left(-\dfrac{x}{4}\right) \cdot \displaystyle\sum_{n=1}^{\infty} (-4)^n \cdot n \cdot x^{n+1}$ //

$= \displaystyle\sum_{n=1}^{\infty} (-4)^{n+1} \cdot n \cdot x^n$ // $-\frac{1}{4} n \cdot (-4x)^n$

$a_n = (-4)^{n+1} \cdot n \cdot x^n$ 이라 하자.

by 비판정법

$\displaystyle\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-4)^n \cdot (n+1) \cdot x^{n+1}}{(-4)^{n+1} \cdot n \cdot x^n} \right| = |-4x|$

① $x = -\frac{1}{4}$

$$\sum_{n=0}^{\infty} -\frac{1}{4}n = -\infty$$

② $x = \frac{1}{4}$

$$\sum_{n=0}^{\infty} -\frac{1}{4}n \cdot (-1)^n$$

by 교급판

$$\lim_{n\to\infty} -\frac{1}{4}n \neq 0 \quad (\text{발산})$$

$\therefore R = \frac{1}{4}, \quad -\frac{1}{4} < x < \frac{1}{4}$

(3) $\frac{x}{(1+4x)^2}$

$= x \cdot (1+4x)^{-2}, \quad s = -2$

$= x\left(1 + \binom{-2}{1}(+4x)^1 + \binom{-2}{2}(4x)^2 + \cdots + \binom{-2}{n}(4x)^n\right)$

$= x\left(1 - 8x + 48x^2 + \cdots + \frac{2\cdot 3 \cdots (2+(n-1))}{n!} \cdot (-1)^n \cdot (4x)^n\right)$

$= x \cdot \sum_{n=0}^{\infty} \frac{1\cdot 2\cdot 3 \cdots (n+1)}{1\cdot n!} \cdot (-1)^n \cdot (4x)^n$

$= \sum_{n=0}^{\infty} (n+1) \cdot (-4)^n \cdot x^{n+1}$ //

$= \sum_{n=1}^{\infty} (-4)^{n-1} \cdot n \cdot x^n$

2.

(1) $f(x) = \ln(1+x)$

$f'(x) = \frac{1}{1+x}$

$= \sum_{n=0}^{\infty} (-x)^n, \quad |x| < 1$ // $(-1)^n \cdot (+)^n$

$\therefore f(x) = \sum_{n=0}^{\infty} \int_0^x (-t)^n dt + f(0)$

$= \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{n+1}}{n+1}$ //

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{n+2}\cdot(n+1)}{(n+2)\cdot x^{n+1}}\right| = |x| < 1$$

$$\therefore \sum_{n=0}^{\infty}(-1)^n\cdot\frac{x^{n+1}}{n+1}, \quad R = 1$$

(2)

$$\frac{\ln(1+x)}{x} = \frac{1}{x}\left(x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots + (-1)^n\cdot\frac{x^{n+1}}{n+1}\right)$$

$$= 1 - \frac{x}{2} + \frac{x^2}{3} + \cdots + (-1)^n\cdot\frac{x^n}{n+1}$$

$$\therefore \int_0^{\frac{1}{2}}\frac{\ln(1+x)}{x}\,dx$$

$$= \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2\cdot 2} + \frac{\left(\frac{1}{2}\right)^3}{3\cdot 3} + \cdots$$

$|S - S_n| < 0.01$ 이므로

$$a_2 = \frac{1}{16} > 0.01, \quad a_3 = \frac{1}{72} > 0.01, \quad a_4 = \frac{1}{216} < 0.01$$

$\therefore n = 3$ //

$$S_3 = \frac{65}{144}$$ //

3. $f(x) = e^x \sin x$

$$e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!}, \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

$$= \sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!}\cdot(-1)^n$$ //

$$\therefore f(x) = \sum_{n=0}^{\infty}\frac{x^n}{n!}\cdot\frac{x^{2n+1}}{(2n+1)!}\cdot(-1)^n$$

$\therefore x^4$의 계수 $a_4 : -\frac{1}{6}$ //